Teacher Model Answers and Marking Rubric

Teacher reference material. Use the exact question IDs and maximum marks.

Q1 — Maximum 5 marks

Question: Explain what a database is and state two benefits of using a database.

Model answer: A database is an organized collection of related data. Benefits include easier retrieval, reduced duplication, improved consistency, controlled access, and reliable sharing.

Marking rubric: Definition of organized related data: 2; One valid benefit: 1; Second distinct valid benefit: 1; Relevance and clarity: 1

Q2(a) — Maximum 3 marks

Question: Define a primary key in a relational database.

Model answer: A primary key is a column or set of columns that uniquely identifies each row in a table. Its values must be unique and cannot be NULL.

Marking rubric: Uniquely identifies a row: 2; Mentions uniqueness and/or non-NULL: 1

Q3 — Maximum 4 marks

Question: What is a foreign key? State its purpose.

Model answer: A foreign key is a column or set of columns that refers to a key in another or the same table. It establishes relationships and helps maintain referential integrity.

Marking rubric: References a key in another/same table: 2; Explains table relationships: 1; Mentions referential integrity: 1

Q4 — Maximum 5 marks

Question: Explain database normalization and give one reason it is useful.

Model answer: Normalization organizes data into related tables to reduce duplication and improve integrity. It can reduce insertion, update, and deletion anomalies.

Marking rubric: Organizes/decomposes data into related tables: 2; Explains reducing redundancy: 1; Valid benefit such as integrity/fewer anomalies: 2

Q5 — Maximum 5 marks

Question: Differentiate between a primary key and a foreign key.

Model answer: A primary key uniquely identifies a row in its own table. A foreign key references a key in another or the same table to create relationships and maintain referential integrity.

Marking rubric: Correct primary-key explanation: 2; Correct foreign-key explanation: 2; Meaningful difference/purpose: 1